Introduction:

Red Dye No.3, formally known as Erythrosine, is a synthetic additive that colours substances a vivid red. It serves no nutritional purpose or preservative function and is used solely for aesthetic appeal.

Erythrosine has been labelled as a known carcinogen in rats; alongside tumour growth, research has also linked food colourants to potential neurodevelopmental issues in children (1,2). Over twenty countries (including the U.S., European Union, Mexico, United Kingdom, Australia, New Zealand) have either banned this substance entirely or have extremely limited permitted use (30)[1].

However, while global regulatory trends shift towards prohibition, Canada permits a broad continued use of this substance (3). As of 2026, Erythrosine remains on the official *List of Permitted Food Colourants* published by Health Canada (3,4).

This paper proposes a strategy blueprint addressed to the Honourable Marjorie Michel, the current Minister of Health, recommending that Erythrosine should be removed from the *List of Permitted Food Colours* (under section B.06.002 of the Food and Drug Regulations) by 2027 (3,4,5). Parallel regulatory action should follow to prohibit its use in pharmaceuticals (under section C.01.040.2(3) of the Food and Drug Regulations) by 2028 (3,4,5)[2].

Background:

In Canada, Erythrosine is found across a variety of food products, including candy, baked goods, ice cream, cereal, fish, and meat[3]. Beyond food, it is found in several medications, including children's cough syrup, allergy medicines, and certain formulations of ibuprofen.

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1 i.e. only in cocktail/maraschino cherries (1)

2 Giving pharmaceutical companies an extra year is a common regulatory courtesy – changing the "inactive ingredients" in a drug requires more stability testing and potential re-approval than changing a recipe for candy.

3 Common brand names including Erythrosine are JellyBelly, Trolli Gummies, Pez Candy, and Double Bubble Bubblegum (45,47).

Erythrosine is regulated by Health Canada by the Bureau of Chemical Safety under the Food and Nutrition Directorate, led by Dr. Joyce Boye.

[4]Under the Food and Drugs Act (R.S.C. 1985, c. F-27, s 4(1)(a-d)), Erythrosine is codified by the Food and Drugs Regulations (C.R.C., c.870) (3,4,5). Specifically, its authorization for food is outlined by section B.06.002 – listed in the *List of Permitted Food Colourants* as entry E.1 – and by C.01.040.2 for pharmaceuticals (3,4,5).

Currently, the Food and Drugs Regulations permit a maximum concentration of 300 mg/kg (ppm) across food categories, with an Acceptable Daily Intake (ADI) of up to 0.1 mg/kg of body weight (3,7,8). For context, for a person weighing 50 kg (110 lbs), the ADI of 0.1 mg/kg would be 50kg x 0.1mg/kg = 5mg per day. Given that some products can contain up to 300mg/kg (ppm), it would only take 16.6 grams of a product dyed at maximum concentration to reach the daily limit.

Regulatory History and the Health Canada Position

In the 1990s, on the basis of research citing tumour growth in rats, an official U.S. proposal to ban Erythrosine from foods and drugs was announced, but deferred due to a lack of clear human-health risk data (1, 7, 35, 41). In 2020, a WHO/JECFA report reassessed the research and pushed for removal, leading to the 2025 FDA final ruling invoking the Delaney Clause to eliminate Erythrosine by 2027 for food products and 2028 for drugs and medications (1, 7, 41). The Delaney Clause is a provision of the U.S. Federal Food, Drug, and Cosmetic Act that prohibits the approval of any food additive found to cause cancer in either humans or animals, regardless of dosage (1, 41).

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4 In 1996-1991, as part of larger food-additive reforms, Erythorsine underwent a series of FAO/WHO Expert Committee on Food Additives (JECFA) reviews, whereby Erythrosine was flagged by JECFA as a potential thyroid-disruptor, leading to the reduction of its original Acceptable Daily Intake (ADI) amount from 0-1.75 mg/kg of bodyweight to an ADI of 0-0.75mg/kg bw 8.In 2018, JECFA again reduced the ADI to 0-0.1mg/kg bw8. Canada incorporated JECFA's findings into its own regulatory policies at each review, meaning that the Canadian judgment on Erythrosine was not independent, but used the same international body that other countries, such as the U.S., were also relying upon (1, 41).

Health Canada acknowledges the FDA's research showing significant correlation between the colourant and tumour growth in rats (1, 21). However, they argue that the weight of evidence is insufficient to establish human harm, and therefore does not warrant restricting its continued use (1, 3, 35). In January 2026, Dr. Maggie Chi, acting on Minister Michel's behalf, responded to a Petition e-6668 that called for stricter regulations of Erythrosine, saying that:

"To date, the weight of scientific evidence has not demonstrated a causal effect or been sufficient to warrant action…Health Canada will continue to monitor emerging scientific developments related to the safety of permitted synthetic colours. If new scientific data becomes available demonstrating that a synthetic colour poses a human health risk…then Health Canada will take action to mitigate that risk, including, if necessary, no longer permitting the ingredient to be used in food and health products" (9). This said, a 2020 Health Canada report stated frankly that Erythrosine, "may pose a risk to human health due to potential carcinogenic effects" (43).

Explaining Current Regulatory Inaction

There are two primary reasons as to why Erythrosine removal is low on Health Canada's political agenda. The first are structural reasons: Erythrosine has not led to any reported deaths, serious illness, or major outbreaks. As such, it is perceived as a low-salience health matter. This, alongside the lack of definitive human harm health-data, provides regulators with rationale for continued inaction (21). In fact, the Department of Health advocated for the use of coloured dyes to "enhance acceptability of medications and compliance, since palatability includes not just taste and smell but also sight to match the medicine's flavour or to conceal unappealing colours" (9).

The second reason is regarding public perception: Health Canada does not want to be viewed as a responder to the FDA; rather, it aims to maintain its status as an independent

scientific review body. They have emphasised that although it coordinates with international regulators, such as the FDA, Health Canada ultimately makes its own decisions based on its own independent regulatory framework and review.

The Asbestos Precedent

Though this paper's argument for the prohibition of Erythrosine is trade-focused, I will briefly mention a parallel case to support my claim that application of the precautionary principle in Canadian food regulation is necessary.

In the 1960s and 1970s, studies showed that asbestos exposure could induce lung tumors in rats (55). In 1977, the IARC concluded there was significant evidence that asbestos was carcinogenic to humans (58). Even so, asbestos continued to be used for decades in insulation and building construction; Canada did not implement federal regulations of asbestos until 2018 (58,59,60).

Current Erythrosine assessments repeat the dangerous pattern: the absence of human-harm data prioritized over conclusive evidence of animal carcinogenicity findings. The issue is not the integrity of the research itself, but rather highlights the systemic failure to apply the precautionary principle when needed. Looking at historical precedents like asbestos, it is clear that we should favor, both ethically and politically, proactive prevention rather than wait for retrospective remediation.

The Policy Window and Multiple Streams Framework Analysis

Kingdon's Multiple Streams Framework (MSF) argues that a policy window is open when the problem, politics, and policy streams converge, making policy change both feasible and politically possible (10).

For Erythrosine, a policy window has opened due to converging factors that have brought these independent streams into alignment.

The Problem Stream:

The first factor is the increased public awareness and corresponding demand for stricter regulation regarding public safety concerns for carcinogenic-causing food substances, a consequence of increased media coverage following the FDA ban (14). Prominent outlets, such as The New York Times, The Atlantic, BBC, CNN, among several others, have heightened public awareness of synthetic food colourants in public discourse; public support for regulatory action is currently strong (14). There is further activation of the Problem Stream through formal petitions such as e-6668, a 2025 citizen-initiated parliamentary petition that demanded stricter legislative action regarding synthetic food additives that gained 525 signatories in one month, validating mounting public pressure (9).

The Politics Stream:

The second and third factors relate to the current economic uncertainty generated by the ongoing U.S.-Canada trade tensions, and the approaching renewal deadline of the Canada-United States-Mexico Agreement (CUSMA/USMCA) in July of 2026 (13).

When the U.S. initiated its International Emergency Economic Powers Act (IEEPA) tariffs against Canada and Mexico, Canada implemented a series of retaliatory tariffs, with the objective to “protect Canadian workers and businesses, and to incentivize a negotiated settlement with the United States”(11). Over the last year, the average U.S. tariff rate has jumped from 2% in 2024 to 16% (11). While a degree of stabilization has occurred, where about 85% of Canada-U.S. trade is currently tariff-free and maintains the lowest average tariff rate among U.S. trade partners, this status remains fragile (11). The July 2026 renewal of the CUSMA will require

Canada to re-enter formal negotiations with Trump's administration, which has already demonstrated willingness and ability to impose sudden and significant tariffs on trade partners.

In this context, removing Erythrosine from Canada's *List of Permitted Food Colourants* presents a low-cost gesture of regulatory alignment with the U.S. that carries meaningful diplomatic value without needing concession on higher-stakes trade issues. Failure to act, however, will maintain a regulatory divergence that may be used as evidence of non-alignment, jeopardizing Canada's current exemption from the IEEPA tariffs under CUSMA. This is a clear activation of Kingdon's Politics Stream, whereby the political environment at large has shifted in a way that makes regulatory alignment strategically advantageous for high-stakes bilateral negotiations (10).

In a press release statement, Prime Minister Carney states that as the government develops new trade relations with the U.S., Canada will "continue to take all measures necessary to support Canadian workers, businesses, and consumers"(11). The removal of Erythrosine represents a high-visibility, actionable, yet politically efficient and low-cost measure that delivers protection for Canadian exporters while strengthening Canada's negotiation position ahead of the CUSMA.

The FDA explicitly states that international food and drug imports to the U.S. must comply with their new compliance requirements (41). As such, if the U.S. and Canada remain misaligned on the regulation of Erythrosine, Canadian companies that manufacture goods containing Erythrosine will face significant operational disruption in January of 2027, when the FDA's compliance deadline for the removal of the substance from food products takes effect. Given that over 70% of Canadian food exports are to the U.S. market, the commercial stakes of regulatory divergence are significant (12). In 2024, over C$800 billion (US$600 billion) in goods

crossed the U.S.-Canada border (12). Canada is also the number one export market for 36 states, and a top-three market for 46 states (12).

Without a federal phase-out, individual Canadian companies that hope to maintain access to the U.S. market will have to unilaterally undertake the associated costs, whether through product reformulation, the creation of separate U.S.-specific production lines, or the acceptance of direct trade loss. In each case, the burden of misalignment, the federal policy gap, falls disproportionately on the Canadian firms, without governmental guidance or support; proactive removal is more administratively efficient than a subsidy or other support mechanism. This is a representation of a second pressure point within the Politics Stream, as the interests of the Canadian exporters, and the government's official commitment to "protect Canadian workers and their businesses" generate direct institutional pressure to act before the 2027 compliance deadline (11).

The Policy Stream:

Finally, the regulatory framework in place is easily modifiable, and politically low-cost and efficient. As aforementioned Erythrosine's permitted use is outlined by the Food and Drugs Regulation (C.R.C., c. 870), which operates under the Food and Drugs Act (R.S.C., 1985, c. F-27) (3,4). Importantly, the removal of Erythrosine does not require a change to the Food and Drugs Act that would require Parliamentary approval. In 1971, parliament already delegated authority to the Cabinet to make direct amendments to the subordinate regulations (6, 15). As such, the entire process is situated within the Executive Branch, requiring no primary legislation and avoiding procedural delays of parliamentary debate or committee review (6,15,16).

Minister Michel maintains the ability to initiate this change through a policy direction, through which the formal regulatory amendment can be made by the Governor in Council

(Cabinet) (6,15,16). The process avoids parliamentary debate and the associated procedural delays and political exposure and risks that may accompany it. This is a clear activation of the Policy Stream, where an administratively accessible solution exists, and the pathway to implementation is open (10).

The Policy Window:

With the problem, politics and policy streams aligned, and the CUSMA July 2026 renewal posing a hard deadline on the policy window which, once closed, may not reopen on favourable terms, the analysis presents Minister Michel with an opportunity for Canada to lead on both public health protection and trade competitiveness in one administratively efficient intervention. The Food, Health and Consumer Products of Canada (FHCP), referenced in a later section, is best positioned to act as the policy entrepreneur that will couple these streams before the CUSMA window closes, having both established ministerial access and direct interest in regulatory alignment (9,12).

Recommendations and Implementation Proposal:

With the policy window open and regulatory pathway established, the following implementation strategy is offered:

Two parallel pathways are proposed: first requiring removal from food products by 2027, and the second requiring removal from pharmaceuticals by 2028. A complete phase-out ban is recommended over mere restriction alone, as partial measures would not resolve regulatory ambiguity. A full ban with a structured timeline would allow companies to coordinate reformulation measures with manufacturers and suppliers, ahead of FDA compliance deadlines. Over the course of the next 12 months, Canadian companies should transition their

Erythrosine-containing products into natural colour alternatives, prioritizing products that are marketed for U.S. distribution and/or marketed for children.

Cabinet Coalition Support

Minister Michel will not act without support. The proposed amendment also has benefits for key Cabinet colleagues. Honourable Maninder Sidhu, the current Minister of Trade, would have direct interest in this amendment for two reasons. First, regulatory alignment with the U.S. would reduce export friction for Canadian manufacturers, as mentioned. Second, entering the CUSMA negotiations with evidence of proactive regulatory alignment would signal Canada's cooperative intentions. Honourable François-Philippe Champagne, the current Minister of Finance, would also have a vested interest in supporting this measure. For Minister Champagne, federal regulatory coordination would be less costly to the Canadian economy, as a harmonized federal phase-out could provide the industry with regulatory certainty, a clear compliance timeline, and companies with the capacity to plan reformulation at scale rather than under duress.

Ministerial Framing and Health Equity Rationale

Furthermore, there are political incentives for Minister Michel to lead this initiative. Minister Michel, publicly known as an advocate for health equity, is well-positioned to frame this intervention as the application of the precautionary principle, directly attempting to protect those most vulnerable to exceeding the ADI through frequent Erythrosine consumption – specifically children and individuals from low-income and racialized communities (48)[5].

Statistically, synthetic colourants such as Erythrosine are found at the highest levels in ultra-processed foods, such as candy and flavoured snacks, which are disproportionately consumed by low-income individuals at increased rates for whom cost, rather than nutritional

[5] The precautionary principle holds that regulatory action is warranted when the potential harm is serious despite scientific uncertainty, a doctrine recognized in Canadian regulatory practice since *114957 Canada Ltée (Spraytech, Société d'arrosage) v. Hudson (Town of)*, 2001 SCC 40 (52).

content, is the primary determinant of food choice (44,48, 49, 50). Individuals from these communities may also lack access to health literacy necessary for awareness over risks associated with the chronic overconsumption of ultra-processed products (44,47). Furthermore, the majority of foods that contain Erythrosine are marketed towards children, who represent the second vulnerable population (46, 51). The substance has appeared in several children's medications, including cough or allergy syrups, as well as in children's nutritional replacements, such as *TruMoo Strawberry Nutrition Shak*e (53, 54). Minister Michel should, as such, position the removal of Erythrosine not as regulatory concession to FDA precedent, but as an evidence-informed decision to protect those who are subject to greater systemic health inequities, consistent with her public profile, supported by Cabinet colleagues, and readily executable through her authority.

Implementation Process:

The implementation process for a complete phased ban as initiated by Minister Michel would be as follows:

1. Before May 2026, Minister Michel should direct Dr. Joyce Boye, the director of Health Canada's Food Directorate, to request her team draft a regulatory amendment that targets section B.06.002 and C.01.040.2 of the Food and Drugs Regulations that currently authorize Erythrosine in foods and drugs, respectively.
2. A Notice of Proposal will notify the Canadian public of Health Canada's intent to amend the Food and Drugs Regulation, providing a summary and rationale for the proposed change.
3. This amendment would be published in the Canada Gazette Part I for a public commentary period. For food additive removals, Health Canada provides stakeholders a

75-day period from the day a Notice of Proposal is published to provide comments. During this period, researchers, scientists, the public at large, alongside any affected industries, such as food manufacturers or pharmaceutical companies, can formally object or voice support.

4. These comments are taken into consideration and responded to in a Notice of Modification (NOM), in which Health Canada will review all submissions and incorporate any changes warranted.
5. If no evidence warrants revision, the Food and Nutrition Directorate will proceed with the NOM acting as the final amended regulation that would be published.
6. Health Canada will then publish the NOM alongside the updated *List of Permitted Food Colourants* in the Canada Gazette Part II, officially specifying the date by removal of Erythrosine is to be enforced.

The parallel drug track governed by C.01.040.2 would follow the same administrative process, with a 2028 deadline for pharmaceutical manufacturers.

Stakeholder Engagement + Communication Strategy

There are additional measures required alongside this amendment initiation to move this proposal forward.

Industry Organization Mobilization

Effective stakeholder mobilization will center on two primary interest groups. First, within the industry sector, the Food, Health and Consumer Products of Canada (FHCP) is a key ally[6]. The FHCP, which represents 119 food, health and consumer product manufacturers,

[6] The FHCP has advocated for increased alignment between U.S. and Canadian trade regulatory policies, supporting the claims that trade divergence creates several barriers for Canadian companies. The FHCP has also recently led discussions with Minister Michel, securing a commitment to prioritize "regulatory modernization," effectively updating the Food and Drugs Regulations to ensure that Canadians have equitable access to safe, high-quality products when needed (9).

actively seeks to reduce trade barriers caused by regulatory divergences with the U.S.(12). They would support the logic for bilateral regulatory coordination and would likely be receptive to the Erythrosine phase-out as an operation that provides trade certainty and elimination of economic harm to Canadian companies. As the policy entrepreneur, the FHCP's public endorsement of the phase-out would signal to Minister Michel that industry support is organized and ready to act.[7]

Ministerial Statement

To ensure the proposal's success, a simultaneous mulit-ministerial communication strategy should be implemented.

The Economic Track: Minister Michel requests colleagues Minister Maninder Sidhu and Minister Champagne to publicly frame the phase-out as a safeguard for Canadian businesses in a time of trade uncertainty.

The Public Health Track: Minister Michel frames the proposal leading with the precautionary principle, referencing the precautionary principle. The proposal centers on protecting the most vulnerable to potential harm from Erythrosine consumption: the children and individuals from racialized, low-income communities (44, 46, 51).

This approach reinforces Health Canada's regulatory autonomy and position as an independent body of scientific review. By framing the ban as a proactive choice with cross-departmental benefits, rather than a belated response to international precedents.

Counterargument: Market-Driven Reformulation

One objection to this proposal warrants direct address. It could be argued that federal intervention is unnecessary, as the market forces may already be driving reformulation: Canadian

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[7] **Health Advocacy Organization Mobilization:** Alongside industry engagement, health advocacy organizations, such as the Canadian Cancer Society could provide public endorsement of the proposal. The Canadian Pediatric Society or the Dietitians of Canada, both of which have issued statements regarding nutrition would be additional sources of stakeholder support. Historically, these organizations have issued statements on child nutrition policies; as children are among the populations most exposed to elevated levels of Erythrosine in both food and drug products, these organizations would affirm the precautionary rationale for the amendment, lend the proposal credibility, and reinforce the health equity framing.

companies, already independently anticipating the consequences of the FDA's regulatory changes, may soon begin (or have begun)[8] voluntary removal of Erythrosine with alternative colourants, accepting the associated costs without governmental oversight. Under this view, regulatory action would be an inefficient duplication of an adjustment that is already underway.

Red Dye 40. Substitution Risk

This objection is not without merit, but it is insufficient on two grounds. First, independent company reformulation does not ensure that the substitutions will be made with public health prioritization as one of the primary criteria. Research indicates that a number of U.S. manufacturers that have chosen to replace Erythrosine with Allura Red AC, commonly known as Red Dye 40, a synthetic petroleum-derived colourant that has been associated with potentially causing neurodevelopmental issues in children (17)[9]. The substitution of one synthetic colourant of concern to another does not represent a public health improvement, especially as new medical research regarding the harmful effects of the colourants comes to light and parents and doctors have increasing concerns about petroleum-based food dyes (17).

Second, and perhaps most importantly, the regulatory landscape surrounding other synthetic dyes, such as Red 40, is already shifting. The FDA is currently working with food manufacturers, retailers, and trade associations to eliminate the six remaining certified food additives, including Red 40, by the end of 2027 (17). In announcing this initiative, FDA Commissioner Marty Makary, MD, MPH, stated that given the "epidemic we face of childhood diabetes, obesity, depression, and ADHD, it's common sense to work together to remove these chemicals in an effort to improve children's healthy eating patterns"(17). Should Canada permit the substitution of Erythrosine with Red Dye 40 or other synthetic colourants of concern without

[8] There is no public data as of yet indicating Canadian companies have already begun this process, but the logic of doing so is foreseeable that the proposal addresses the substitution risk nonetheless.
[9] Including neurodevelopmental conditions such as ADHD, anxiety, depression, and diseases such as childhood obesity and diabetes (3)

restriction, Health Canada risks returning to square one, responding to a U.S. ban retroactively, rather than acting in advance of it.

Additional Implementation Requirement

This proposal therefore specifies that when substituting Erythrosine in reformulations, manufacturers must transition into commercially available natural alternatives, such as beetroot extract, paprika extract, turmeric, annatto, and similar fruit-derived colourants, rather than to other synthetic dyes. Instead of possibly responding to successive FDA bans retroactively, Health Canada should position itself as a proactive leader by anticipating the trajectory of international food safety regulation as findings develop. By restricting substitution to natural alternatives, the measure will also assist in the reformulation burden Canadian companies may face in the future as the FDA moves towards comprehensive synthetic colourant restrictions. Canadian federal oversight is necessary, even if companies would have reformulated without governmental direction.

Bibliography

1. *U.S. Food and Drug Administration. (2025, January 15). FD&C Red No. 3. https://www.fda.gov/industry/color-additives/fdc-red-no-3*
2. *California Office of Environmental Health Hazard Assessment. (2021, April 16). Health effects assessment: Potential neurobehavioral effects of synthetic food dyes in children. https://oehha.ca.gov/risk-assessment/report/health-effects-assessment-potential-neurobehavioral-effects-synthetic-food-dyes-children*
3. *Health Canada. (2025, October 20). List of permitted food colours (lists of permitted food additives). Government of Canada. https://www.canada.ca/en/health-canada/services/food-nutrition/food-safety/food-additives/lists-permitted/3-colouring-agents.html*
4. *Justice Laws Website. (n.d.). Food and Drug Regulations, C.R.C., c. 870, s. C.01.040.2. Government of Canada. https://laws-lois.justice.gc.ca/eng/regulations/C.R.C.%2C_c._870/section-C.01.040.2.html*
5. *Justice Laws Website. (n.d.). Food and Drugs Act (R.S.C., 1985, c. F-27), s. 4. Government of Canada. https://laws-lois.justice.gc.ca/eng/acts/F-27/section-4.html*
6. *Justice Laws Website. (n.d.). Food and Drugs Act (R.S.C., 1985, c. F-27), s. 30. Government of Canada. https://laws-lois.justice.gc.ca/eng/acts/F-27/section-30.html*
7. *World Health Organization & Food and Agriculture Organization. (n.d.). Erythrosine. JECFA Database. https://apps.who.int/food-additives-contaminants-jecfa-database/Home/Chemical/3740*
8. *Food and Agriculture Organization & World Health Organization. (2018). Summary and conclusions of the eighty-sixth meeting of the Joint FAO/WHO Expert Committee on Food Additives. https://www.fao.org/fileadmin/user_upload/agns/pdf/jecfa/JECFA86_Summary_Report.pdf*
9. *House of Commons of Canada. (2026, January 26). Petition e-6668. https://www.ourcommons.ca/petitions/en/Petition/Details?Petition=e-6668*
10. *Kingdon, J. W. (1995). Agendas, alternatives, and public policies (2nd ed.). HarperCollins College.*
11. *Prime Minister of Canada. (2025, August 22). Statement by the Prime Minister on CAN-U.S. trade. https://www.pm.gc.ca/en/news/statements/2025/08/22/statement-prime-minister-can-us*
12. *Food, Health & Consumer Products of Canada. (n.d.). Canada–U.S. trade relations. https://www.fhcp.ca/en/Advocacy/CanadaUS-Trade-Relations*
13. *Government of Canada. (2025). CUSMA review — question period notes. Open Government. https://search.open.canada.ca/qpnotes/record/dfatd-maecd%2C00006-2025*
14. *Florko, N. (2025, April 22). RFK Jr. is cracking down on food dyes. The Atlantic. https://www.theatlantic.com/health/archive/2025/04/food-dye-crackdown-rfk-jr/682548/*
15. *Health Canada. (n.d.). Modification to the lists of permitted food additives as part of the modernization of Part B of the Food and Drug Regulations. Government of Canada. https://www.canada.ca/en/health-canada/services/food-nutrition/legislation-guidelines/acts-regulations/notices-proposal-notices-modification/lists-permitted-food-additives-modernization-part-b-food-drug-regulations.html*
16. *Health Canada. (n.d.). Transition guide: Understanding and using the lists of permitted food additives. Government of Canada. https://www.canada.ca/content/dam/hc-sc/migration/hc-sc/fn-an/alt_formats/pdf/securit/addit/list/transition-eng.pdf*
17. *U.S. Food and Drug Administration. (2026, March 25). Tracking food industry pledges to remove petroleum-based food dyes. https://www.fda.gov/food/color-additives-information-consumers/tracking-food-industry-pledges-remove-petroleum-based-food-dyes*
18. *Abacus Data. (2025). Public support for government action on plastics and toxics. Environmental Defence. https://environmentaldefence.ca/wp-content/uploads/2025/04/EDC-Omnibus-Research-Questions-4.22.25-Plastics-Toxics.pdf*
19. *Anwar, F., Singh, R., & Mushtaq, G. (2015). Cancer initiating properties of erythrosine supplements with sub necrotic dose of diethyl nitrosamine: Potential effects on biochemical parameters of liver, vitamin C and E. Molecular & Cellular Toxicology, 11, 357–366. https://doi.org/10.1007/s13273-015-0036-0*
20. *Borzelleca, J. F., Capen, C. C., & Hallagan, J. B. (1987). Lifetime toxicity/carcinogenicity study of FD&C Red No. 3 (erythrosine) in rats. Food and Chemical Toxicology, 25(10), 723–733. https://doi.org/10.1016/0278-6915(87)90226-2*
21. *Borzelleca, J. F., & Hallagan, J. B. (1987). Lifetime toxicity/carcinogenicity study of FD&C Red No. 3 (erythrosine) in mice. Food and Chemical Toxicology, 25(10), 735–737. https://doi.org/10.1016/0278-6915(87)90227-4*
22. *California Environmental Protection Agency, Office of Environmental Health Hazard Assessment. (1999). Final prioritized candidate chemicals under consideration for carcinogenicity evaluation: Fifty-four chemicals within Batch #3. https://oehha.ca.gov/media/downloads/crnr/fbatch3.pdf*
23. *California Environmental Protection Agency, Office of Environmental Health Hazard Assessment. (n.d.). Synthetic food dye risk assessment. https://oehha.ca.gov/risk-assessment/synthetic-food-dye-risk-assessment*
24. *Canada Border Services Agency. (n.d.). Canada tariff finder. https://www.tariffinder.ca/en/getStarted*
25. *Canada Gazette. (2018, January 6). Prohibition of asbestos and products containing asbestos regulations. Canada Gazette, Part I. https://gazette.gc.ca/rp-pr/p1/2018/2018-01-06/html/reg3-eng.html*

26. Center for Science in the Public Interest. (2022). Petition to the Food and Drug Administration to delist FD&C Red No. 3. https://www.cspi.org/sites/default/files/2022-10/Red%203%20petition_24%20Oct%202022_FINAL%20%281%29.pdf
27. Chequer, F. M., Venâncio, V. P., & Almeida, M. R. (2017). Erythrosine B and quinoline yellow dyes regulate DNA repair gene expression in human HepG2 cells. Toxicology and Industrial Health, 33(10), 765–774. https://doi.org/10.1177/0748233717715186
28. Dees, C., Askari, M., & Garrett, S. (1997). Estrogenic and DNA-damaging activity of Red No. 3 in human breast cancer cells. Environmental Health Perspectives, 105(Suppl. 3), 625–632. https://doi.org/10.1289/ehp.97105s362
29. Department of Finance Canada. (n.d.). Canada's response to U.S. tariffs. Government of Canada. https://www.canada.ca/en/department-finance/programs/international-trade-finance-policy/canadas-response-us-tariffs.html
30. European Food Safety Authority. (2011). Scientific opinion on re-evaluating erythrosine (E 127) as a food additive. EFSA Journal, 9(1), Article 1854. https://efsa.onlinelibrary.wiley.com/doi/pdf/10.2903/j.efsa.2011.1854
31. Fassler, A. (2023, May 15). The FDA's artificial-food-dye problem. The Atlantic. https://www.theatlantic.com/health/archive/2023/05/artificial-food-dye-red-no-3-fda/674099/
32. Ganesan, L., Margolles-Clark, E., Song, Y., & Buchwald, P. (2011). The food colorant erythrosine is a promiscuous protein-protein interaction inhibitor. Biochemical Pharmacology, 81(6), 810–818. https://doi.org/10.1016/j.bcp.2010.12.020
33. Global Affairs Canada. (n.d.). Canada-United States engagement. Government of Canada. https://international.canada.ca/en/global-affairs/campaigns/canada-us-engagement
34. Health Canada. (2023). Notices of proposal and notices of modification for food additives. Government of Canada. https://www.canada.ca/en/health-canada/services/food-nutrition/legislation-guidelines/acts-regulations/notices-proposal-notices-modification.html#a1
35. Health Canada. (n.d.). Safety of food additives. Government of Canada. https://www.canada.ca/en/health-canada/services/healthy-living/your-health/food-nutrition/safety-food-additives.html
36. Justice Laws Website. (n.d.). Food and Drug Regulations (C.R.C., c. 870), section B.06.002. Government of Canada. https://laws-lois.justice.gc.ca/eng/regulations/c.r.c.,_c._870/section-B.06.002-20070419.html
37. Public Works and Government Services Canada. (2023). Canada Gazette, Part I, 157(44). https://gazette.gc.ca/rp-pr/p1/2023/2023-11-04/pdf/g1-15744.pdf
38. Schwarcz, J. (n.d.). Red dye 3 isn't a felon lurking in our food supply. McGill Office for Science and Society. https://www.mcgill.ca/oss/article/critical-thinking-health-and-nutrition/red-dye-3-isnt-felon-lurking-food-supply
39. Trade Commissioner Service. (n.d.). Understanding CUSMA compliance. Government of Canada. https://www.tradecommissioner.gc.ca/en/market-industry-info/search-country-region/country/canada-united-states-export/us-tariffs/understanding-cusma-compliance.html
40. U.S. Customs and Border Protection. (n.d.). United States-Mexico-Canada Agreement (USMCA). https://www.cbp.gov/trade/priority-issues/trade-agreements/USMCA
41. U.S. Food and Drug Administration. (2025, January 16). Color additive petition from Center for Science in the Public Interest, et al.; request to revoke color additive listing for use of FD&C Red No. 3 in food and ingested drugs. Federal Register.https://www.federalregister.gov/documents/2025/01/16/2025-00830/color-additive-petition-from-center-for-science-in-the-public-interest-et-al-request-to-revoke-color https://www.fda.gov/food/hfp-constituent-updates/fda-revoke-authorization-use-red-no-3-food-and-ingested-drugs
42. World Trade Organization. (n.d.). DS135: European Communities — Measures affecting asbestos and products containing asbestos. https://www.wto.org/english/tratop_e/dispu_e/cases_e/ds135_e.htm
43. Health Canada. (n.d.). Pigment red 3: Batch 3 challenge. Government of Canada. https://www.canada.ca/en/health-canada/services/chemical-substances/challenge/batch-3/pigment-red-3.html
44. Mengi Çelik, Ö., Güler, Ü., Ekici, E. M., & Aytekin Şahin, G. (2025). Factors associated with ultra-processed foods consumption: Cooking and food preparation skills and food literacy. BMC Public Health, 25(1), Article 3078. https://doi.org/10.1186/s12889-025-23196-z
45. Cleveland Clinic. (n.d.). Red dye 3: What it is and why it's being banned. https://health.clevelandclinic.org/red-dye-3
46. Batada, A., & Jacobson, M. F. (2016). Prevalence of artificial food colors in grocery store products marketed to children. Clinical Pediatrics, 55(12), 1126–1130. https://doi.org/10.1177/0009922816651621
47. Consavage Stanley, K., Harrigan, P. B., Serrano, E. L., & Kraak, V. I. (2022). A systematic scoping review of the literacy literature to develop a digital food and nutrition literacy model for low-income adults to make healthy choices in the online food retail ecosystem to reduce obesity risk. Obesity Reviews, 23(4), Article e13414. https://doi.org/10.1111/obr.13414
48. Darmon, N., & Drewnowski, A. (2015). Contribution of food prices and diet cost to socioeconomic disparities in diet quality and health: A systematic review and analysis. Nutrition Reviews, 73(10), 643–660. https://doi.org/10.1093/nutrit/nuv027
49. Dicken, S. J., Qamar, S., & Batterham, R. L. (2024). Who consumes ultra-processed food? A systematic review of sociodemographic determinants of ultra-processed food consumption from nationally representative samples. Nutrition Research Reviews, 37(2). https://doi.org/10.1017/S0954422423000240

50. *Dunford, E. K., Galligan, T. M., Taillie, L. S., & Musicus, A. A. (2025). All the colors of the rainbow: Synthetic dyes in US packaged foods and beverages in 2020. Journal of the Academy of Nutrition and Dietetics, 125(9), 1207–1217. https://doi.org/10.1016/j.jand.2025.05.007*
51. *Lehmkuhler, A. L., Kruzliakova, N., Hollingsworth, D. W., Lee, M. M., Camden, M. C., Mitchell, A. E., Jones, A. D., & Kraak, V. I. (2020). Certified food dyes in over-the-counter medicines and supplements marketed for children and pregnant women. Food and Chemical Toxicology, 143, Article 111499. https://doi.org/10.1016/j.fct.2020.111499*
52. *114957 Canada Ltée (Spraytech, Société d'arrosage) v. Hudson (Town of), 2001 SCC 40. https://decisions.scc-csc.ca/scc-csc/scc-csc/en/item/1878/index.do*
53. *TruMoo. (n.d.). Strawberry whole milk quart. https://trumoo.com/products/strawberry-whole-milk-plastic-quart/*
54. *Lehmkuhler, A. L., Miller, M. D., Bradman, A., Castorina, R., & Mitchell, A. E. (2020). Certified food dyes in over-the-counter medicines and supplements marketed for children and pregnant women. Food and Chemical Toxicology, 143, Article 111499. https://doi.org/10.1016/j.fct.2020.111499*
55. *Gross, P., de Treville, R. T. P., Tolker, E. B., Kaschak, M., & Babyak, M. A. (1967). Experimental asbestosis: The development of lung cancer in rats with pulmonary deposits of chrysotile asbestos dust. Archives of Environmental Health, 15(3), 343–355. https://doi.org/10.1080/00039896.1967.10665045*
56. *Wagner, J. C., Sleggs, C. A., & Marchand, P. (1960). Diffuse pleural mesothelioma and asbestos exposure in the North Western Cape Province. British Journal of Industrial Medicine, 17(4), 260–271. https://doi.org/10.1136/oem.17.4.260*
57. *Selikoff, I. J., Churg, J., & Hammond, E. C. (1964). Asbestos exposure and neoplasia. JAMA, 188(1), 22–26. https://doi.org/10.1001/jama.1964.03060270028006*
58. *International Agency for Research on Cancer. (1977). Asbestos (IARC Monographs on the Evaluation of Carcinogenic Risks of Chemicals to Man, Vol. 14). IARC.*
59. *Health Canada. (2023, February 10). Asbestos and your health. Government of Canada. https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/health-risks-asbestos.html*
60. *Government of Canada. (2018). Prohibition of Asbestos and Products Containing Asbestos Regulations, SOR/2018-196. https://laws-lois.justice.gc.ca/eng/regulations/SOR-2018-196/FullText.html*